

Isometric Projection

Isometric projection is a pictorial projection of an object in which a three-dimensional view of an object is shown on a two-dimensional drawing sheet. This projection shows view of three faces of the object equally and hence, it is helpful even to a layman for the proper understanding of the shape of object. It is used by construction engineers for the preparation of pictorial views at the work site as well as by design engineers in the design and development of new or complicated parts, when the shape is difficult to understand from the multiview projection. Three-dimensional piping network can be easily represented by isometric projection. Isometric projection is found more suitable for getting pictorial views of comparatively small objects, because the perspective effect is not considered here.

16.1 PRINCIPLE OF ISOMETRIC PROJECTION

Isometric projection is one of the axonometric projections as explained in Section 9.1 of Chapter 9. It is a pictorial orthographic projection of an object in which a transparent cube containing the object is tilted until one of the solid diagonals of the cube becomes perpendicular to the vertical (picture) plane and the three axes of the cube are equally inclined ($35^{\circ}16'$) to this vertical plane.

Isometric projection of a cube can be theoretically obtained by employing the change of position method and is

shown in Fig. 16.1. Here, the third elevation $a'b'c'd'e'f'g'h'$ is the isometric projection of the cube. The front view of the cube is resting on one of its corners (g') on the ground with a solid diagonal $e'e'$ perpendicular to VP. This view (front view) shows the right, top and left square faces of the cube as rhombus with equal measure "*Isometric*". It is to be noted that the three faces together form a regular hexagon with inclined edges at 30° to horizontal. The isometric projection of the cube is alone reproduced in Fig. 16.2. In order to distinguish the isometric projection from the usual front view, capital letters are used in this book.

When the three axes (the three mutually perpendicular edges $c'a'$, $c'f'$, and $c'h'$ of the cube in the front view of Fig. 16.1) are equally inclined $35^{\circ}16'$ to the vertical (picture) plane, the edges $g'f'$ and $g'h'$ are seen 30° inclined to the reference line, while the $g'e'$ is seen vertical. Since the edges are equally inclined to VP, they are equally foreshortened to a value of cosine of $35^{\circ}16'$ (approx. 82%). The pictorial view formed by isometric projection can be drawn directly from the projections of the solid in simple position. Figures 16.3 and 16.4 show the method applied to a cube and a rectangular prism. Here, the x and y direction measurements are marked at 30° to the horizontal towards right and left respectively, while the z direction vertical upwards as shown. The edge lengths are marked along these axes after foreshortening to 82%.

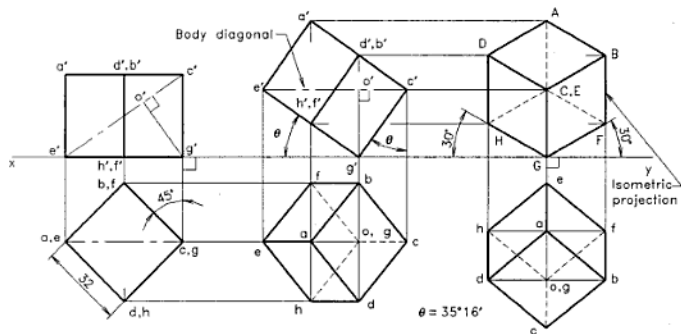


Fig. 16.1 Isometric projection from conventional orthographic projections.

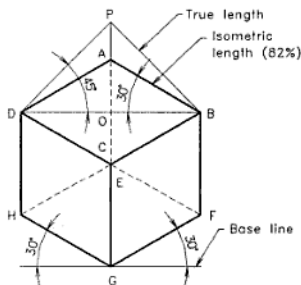


Fig. 16.2 Isometric projection from conventional orthographic projections.

16.2 ISOMETRIC SCALE

In the isometric projection of a cube shown in Fig. 16.2, the top face ABCD is sloping away from the observer and hence the edges of the top face will appear foreshortened. The true shape of the triangle DAB is represented by the triangle DPB.

The extent of reduction of an isometric line can be easily found by constructing a diagram called *isometric scale*. For this, reproduce the triangle DPA as shown in Fig. 16.5. Mark the division of true length on DP. Through these divisions draw vertical lines to get the corresponding points on D'A'.

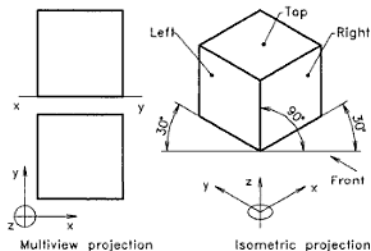


Fig. 16.3 A cube.

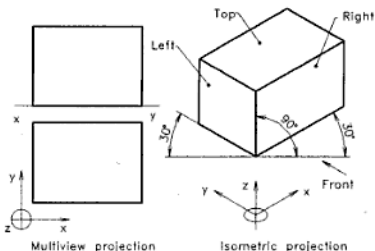


Fig. 16.4 A rectangular prism.

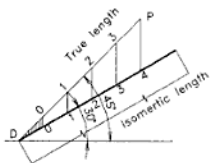


Fig. 16.5 Construction of isometric scale.

The divisions of the line DA give the dimensions to isometric scale.

From the triangles ADO and PDO, the ratio of the isometric length to the true length.

$$\begin{aligned} \text{i.e. } DA/DP &= \cos 45^\circ / \cos 30^\circ \\ &= 0.816 \text{ (cosine of } 35^\circ 16') \end{aligned}$$

The isometric axes are reduced in the ratio

1: 0.816, i.e. 82% approximately.

The isometric scale can be drawn in a simple form as shown in Fig. 16.6. The true length is marked as AB. A line AC is drawn at 15° to AB to intersect another line inclined at 45° to

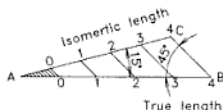
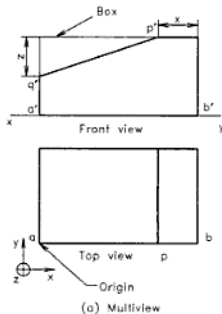
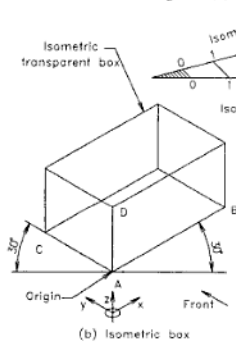


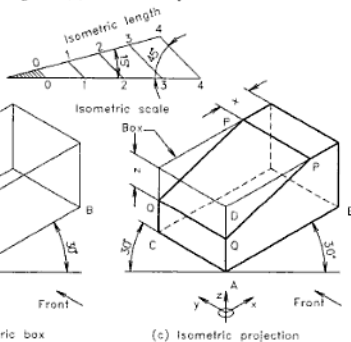
Fig. 16.6 Simplified isometric scale.



(a) Multiview



(b) Isometric box



(c) Isometric projection

Fig. 16.7 Isometric projection (box method).

horizontal and drawn from B. Then 45° inclined lines are drawn from the divisions on the true length scale AB, so that the corresponding isometric lengths are obtained on the line AC. Even though it is easy to calculate the isometric lengths (82%) with the help of a calculator, it is a custom to show isometric scale nearby the isometric projection (on top or right side) for identification.

16.3 PROCEDURE FOR DRAWING ISOMETRIC PROJECTION

Even though the isometric projection of an object can be drawn by ordinary change of position of view method or auxiliary projection method, they are time consuming and tedious. With the help of the top and front views in simple position, the isometric projection of an object can be directly drawn by using any one or a combination of the two following methods:

1. Box method
2. Coordinate or offset method.

Box Method

In this method, the object is assumed to be enclosed in a transparent rectangular box of size just to fit the object, at the same time the sides of the box are parallel to the three reference planes. This transparent box is drawn first, over the top and front views using thin lines.

The zero position of the x, y and z axes of measurements and their +ve directions are marked in the top view as given in Fig. 17.7(a). Let the zero position be at the corner a (the

'origin'). The circle with '+' mark may be assumed the upward direction from the plane of paper.

After drawing an isometric scale, the isometric drawing can be started. From the orthographic views, take the dimensions in the x , y , z directions from the 'origin' a , convert them into isometric lengths (either by using isometric scale or by using calculator) and draw the transparent box [Fig. 16.7(b)]. For this, draw a horizontal reference line of short length and mark point A (origin) at the middle of it. Draw two 30° inclined lines and a vertical line through the point A to locate the three isometric axes AB, AC, and AD respectively to represent the mutually perpendicular edges of the box in isometric projection.

After completing the box, the object is constructed inside, relative to the edges of the box (see Fig. 16.8(c)). On the isometric box the line PP can be marked by locating distance ' x ' multiplied by 0.82 from the end face of box. Similarly, the line QQ can be marked from top face at ' z ' distance in isometric length. By marking various points in the respective xyz directions using isometric lengths and converting the thin lines representing the object to the visible and hidden edges, the isometric drawing is completed.

Coordinate or Offset Method

In this method, one of the three isometric planes (top, left or right) is taken as the reference plane and the end points of the edges of the object are marked along an axis perpendicular to this reference plane. Figure 16.8 shows the construction of

the isometric drawing by coordinate method. After drawing the top and front views and the isometric scale, the construction of the isometric projection is started by drawing the base of the object. As is done in the box method, the horizontal reference line of short length is drawn and the zero position of the axes (point A) is marked at the middle of it. Now, the 30° lines are drawn to locate the isometric x and y axes. The base of the object is now completed in the x and y directions giving the isometric rectangle. The vertical height points are then marked by measuring the z values from the front view and converting them to the isometric lengths. For example, to get point p , the height z is marked at x distance along the x axis. After marking all the required points and converting the thin lines representing the object to the visible and hidden edges, the isometric drawing is completed.

Coordinate or offset method is best suited for objects containing a large number of non-isometric lines and planes. For certain shapes, a combination of the two methods may have to be applied to get the drawing in a shorter time. It is to be noted that, in isometric projection all the measurements are taken in the x , y , z directions only and they are marked along the respective isometric axes after multiplying by the isometric scale factor 0.82.

Hidden Lines

The hidden edges of the isometric projections are represented by short dashes (Type E or F) and the rules for drawing them are the same as followed for multiview projection. Hidden

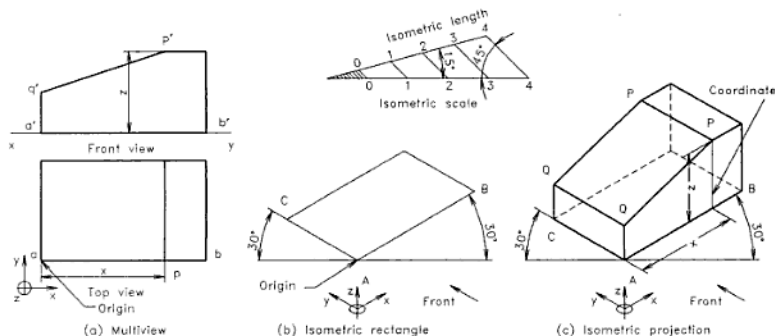


Fig. 16.8 Isometric projection (coordinate method).

lines are generally omitted in isometric projection of objects having a number of hidden edges, because drawing of these lines will reduce the clarity. But simple objects as well as objects requiring the drawing of hidden details, are provided with the invisible informations of the shape.

Selection of Isometric Axes

The isometric axes may be placed in any desired position with respect to the objects. Generally, the axes are determined by the position from which the object is usually viewed or by the position which describes the shape of object more clearly. Figure 16.9 shows the result of varying the origin of the axes and their directions. Here, the second isometric view, showing the top side and keeping the origin at B, gives a better clarity to the shape. To show the bottom side of the object, the third or fourth position may be selected.

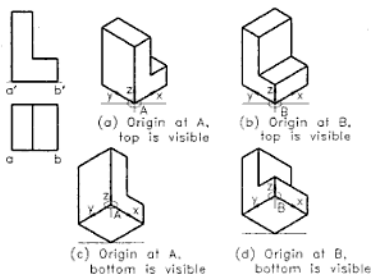


Fig. 16.9 Isometric projection of various axis positions and directions.

Dimensioning

The general rules for the dimensioning of multiview projection is applicable for isometric projection also, except the following.

1. All the extension lines and dimension lines should be parallel to the isometric axes and they should lie on any of the isometric planes.
2. The text should be placed at the middle of the dimension line, after breaking it for a short length.
3. The dimensional values in x direction should be readable from the right side, while those in y direction should be readable from the left side. The dimensional values in z direction should be readable horizontally from the right side.

4. The numerals placed along the three axes should be aligned with the direction of the axes.

Figure 16.10 shows the recommended dimensioning layout for isometric drawing by B.I.S.

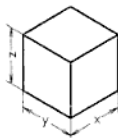


Fig. 16.10 Dimensioning of isometric drawings.

16.4 ISOMETRIC PROJECTION AND ISOMETRIC VIEW

As discussed earlier, isometric projection of an object is the front view of the object placed in the isometric position. Isometric projection is the actual projection of the object on VP. Here, as the edges of the transparent cube are inclined $35^{\circ}16'$ to VP, their projection on VP will have a length of about 82% of the true length, when measured in the isometric position.

To avoid the difficulty in determining the foreshortened lengths, the foreshortening of the axes may be ignored. Hence, isometric projection can be drawn directly, using the true length of the edges of the cube along the isometric axes. As a result, the projection obtained is larger in size than the actual. This projection is called *isometric view* or *isometric drawing*. Thus in short, as the length of the edge of the cube, which are inclined to VP, are taken as equal to the true length of the object itself, the view obtained will be larger than the isometric projection. This enlargement will be to the tune of $1/0.816$ (i.e. 22.5% larger).

An isometric projection and an isometric view of a cube are shown in Fig. 16.11.

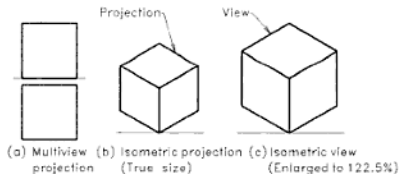


Fig. 16.11 Comparison between isometric projection and isometric view.

Notes:

1. An isometric projection of the x , y , z axes are equally inclined to the vertical plane, hence the edges of the object which are parallel to them are foreshortened (to about 82%).
2. The three isometric axes form the three isometric planes namely left, right and top.
3. All the dimensions are measured and marked along the isometric axes only.
4. Isometric projection is the actual size of the pictorial view after foreshortening the true length to about 82%. The isometric scale is drawn nearby the figure to identify the isometric projection.
5. A line or plane which is not parallel to the three axes or the three planes, is called a non-isometric line or plane.
6. The location of the origin of axes may be suitably selected to show maximum details clearly. Generally the front side may be taken towards left or towards right, suitable to the object.
7. Drawing of the isometric projection or view may be done using box method, coordinate (offset) method or a combination of the two.
8. The hidden edges are shown usually for simple objects only or for indicating certain important hidden details.
9. The edge of isometric projection or view are named using capital letters.
10. Isometric drawings are dimensioned along the three axes only.
11. The dimensional value is entered at the middle of the dimension line, after breaking it for a short length. The text should be entered along the directions of x and y . For z direction it should be readable from right side.

16.5 PLANE FIGURES

Isometric projection or view of plane figures are generally drawn on any one of the isometric planes such as top, right or left.

Pentagonal Lamina

Figure 16.12 shows isometric projection of a pentagonal lamina. Here the lamina is kept parallel to HP so that the top isometric plane is seen. To get the projection, the top view of lamina is enclosed in a rectangle 1234 and the origin is marked. Draw a reference line and mark point 1 at the middle of it. Construct the isometric rectangle 1234, parallel to the isometric axes x and y , after converting the side length as per

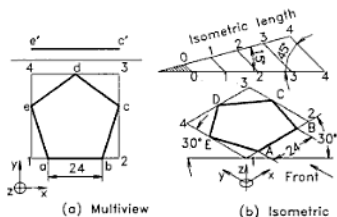


Fig. 16.12 A pentagonal lamina.

the isometric scale. To mark point A, measure the distance in the top view from 1 to a, convert that to isometric length and mark it along the isometric x -axis. Similarly, locate the remaining points BCDE on the isometric rectangle. Join the points by thick line to complete the isometric projection on the top plane.

Circular Lamina

Figure 16.14 shows isometric views of a circular lamina of 50 mm diameter, seen on left, top and right planes as ellipses. To get the views, enclose the given circle in a square 1234.

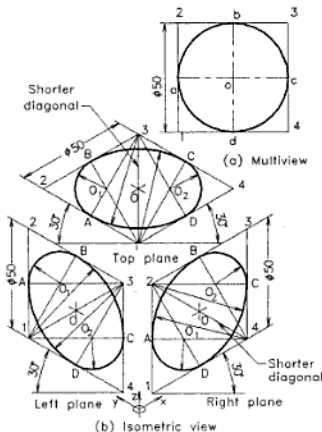


Fig. 16.13 A circular lamina (four-centre construction).

Construct three isometric squares (rhombus) of 50 mm side for the left, top, and right faces as shown in figure. Since the isometric view (not the projection) is required, there is no need of drawing the isometric scale and foreshorten the lengths. In an isometric view, a circle is seen as an isometric circle, i.e. ellipse.

The isometric circle (ellipse) may be constructed approximately by arcs using the four centre method. To locate the four centres and the ends of arcs, join the mid points A, D to the corner 3 and B, C to the corner 1 of the shorter diagonal in the top isometric plane. The intersection points O_1 and O_2 are the centres for short arcs, while the corners 1 and 3 are the centres for long arcs. Draw the short arcs AB and DC with centres O_1 and O_2 respectively. Similarly, draw long arcs BC and DA with centres 1 and 3 respectively to complete the approximate ellipse. The same is repeated on the other two faces to get the isocircles.

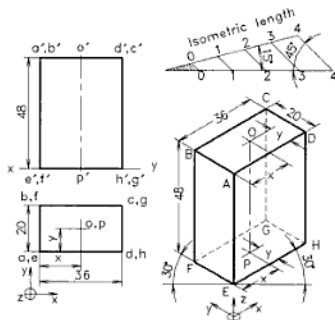


Fig. 16.14 Isometric projection.

16.6 SOLIDS

Isometric projection or view of a solid can be drawn by following the box method or coordinate method explained earlier. The isometric circles may be easily drawn by using the four centre method. The following examples explain the procedure of drawing.

Example 16.1

Draw the isometric projection of a rectangular prism of side of base 36 mm $\times$ 20 mm and height 48 mm, resting upon its base on HP and the 36 mm long edges are parallel to VP.

Refer to Fig. 16.14.

1. Draw multi-view projection of the prism in the given position and name the corners. Locate the origin of axes on the left side of top view in order to get the front view on the right isometric plane.
2. Construct the isometric scale as explained earlier (Fig. 16.6) on the top or right side of the area provided for the isometric projection.
3. Draw a short horizontal thin line and mark the mid point of it as the origin (corner E) of the isometric axes xyz . From the origin draw 30° inclined line EH towards right to represent x direction and another 30° line EF towards left to represent y direction. The vertical line EA upwards represents the z axis. A symbol for isometric axes may also be marked below this origin as shown in figure for identifying the directions without mistakes.
4. From the top view take the distance eh along x direction, multiply it by 0.82 and mark it along the x isometric axis as EH. Again from the top view

take the distance ef along y direction, multiply it by 0.82 and mark it along the y isometric axis as EF. Similarly, from the front view take the distance $e'a'$ along z direction, multiply it by 0.82 and mark it along the z isometric axis as EA. Repeat the same for all edges along xyz directions to get the isometric projection, drawn in thin lines.

5. To locate the vertical axis OP, measure the x and y distances of o, p from the top view, multiply it by 0.82 and mark it along the x and y isometric axes from the corner E at the bottom, as well as from A at the top corner of the isometric box. Join OP by chain line along the z axis to indicate the axis.
6. Convert the thin lines to visible and hidden edges as done for projections of solids. Name the corners and axis using capitals, and print the given dimensions as permitted for pictorial views (Refer Fig. 16.10) to complete the drawing.

Example 16.2

Draw isometric view of a hexagonal prism of 50 mm height and side 20 mm long, lying on HP with the axis perpendicular to VP. Select the origin of the isometric axes suitable to get the front view on the left isometric plane.

Refer to Fig. 16.15.

1. Draw the multiview projection of the prism in the given position.
2. Enclose the front view in a box $1'2'3'4'$ and locate the origin of the isometric axes on the right corner of the top view in order to get the front view on the left (yz) isometric plane as shown.

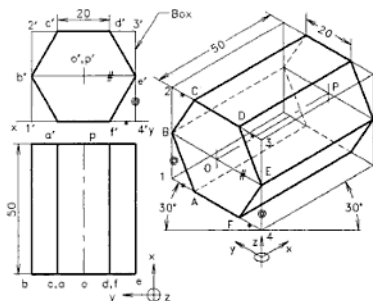


Fig. 16.15 Isometric view (box method).

3. Draw a short horizontal thin line and mark the mid point of it as the origin (4) of the isometric axes xyz . From the origin (4) draw 30° inclined thin lines towards right and left directions to represent x and y isometric axes respectively and a vertical thin line for z axis. The symbol for isometric axes may also be marked below this origin.
4. Draw the isometric view of the box 1234 using the lengths measured along xyz directions from the front and top views as explained in Example 16.1. Here, the lengths are not foreshortened, because requirement is not an isometric projection but a view.
5. Construct the isometric hexagon ABCDEF on the left isometric plane (1234) of the box. To mark the corner F, take the distance $4'f'$ using a bow divider and mark it along the same (y) direction from 4 to F on the isometric box. Similarly to mark the corner E, take the distance $4'e'$ using a bow divider and mark it along the same (z) direction from 4 to E on the isometric box. Following the same method mark the remaining corners of the hexagon at the relative positions to the box corners 1234 along the x , y and z directions. Repeat the same for the back hexagonal face of the prism. Complete the view by joining the front and back corners with thin lines along the x direction.
6. To locate the horizontal axis OP, join BE, and mark O the mid point as one end of axis. Similarly, on the back face locate P and join OP by chain line to indicate the axis.

7. Convert the thin lines to visible and hidden edges as done for projections of solids. Name the corners and axis using capitals, and print the given dimensions as permitted for pictorial views (Refer Fig. 16.10) to complete the drawing.

Example 16.3

Draw isometric view of a cylinder of 50 mm height and diameter 40 mm, lying on one of its generators on HP with the axis perpendicular to VP. Select the origin of the isometric axes suitable to get the front view on the left isometric plane.

Refer to Fig. 16.16.

1. Draw the multiview projection of the cylinder in the given position.
2. Enclose the front view in a square box $1'2'3'4'$ and locate the origin of the isometric axes on the right corner of the top view in order to get the front view on the left (yz) isometric plane as shown.
3. Draw a short horizontal thin line and mark the mid point of it as the origin (4) of the isometric axes xyz . From the origin (4) draw 30° inclined thin lines towards right and left directions to represent x and y isometric axes respectively and a vertical thin line for z axis. The symbol for isometric axes may also be marked below this origin.
4. Draw the isometric view of the box 1234 using the lengths measured along xyz directions from the front and top views. Here, the lengths are not foreshortened, because requirement is not an isometric projection but a view.

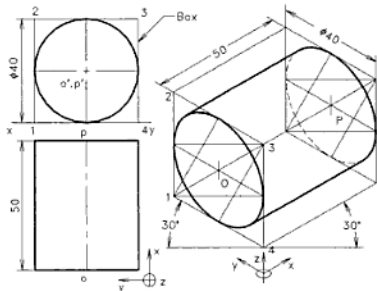


Fig. 16.16 Isometric view (box method).

- Construct the isometric circle (four-centered ellipse) on the left isometric plane (1234) of the box as explained in Fig. 16.13. Repeat the same for the back end face of the box. Complete the view by drawing tangents to the front and back ellipses with thick lines along the x direction.
- To locate the horizontal axis OP , join AC , and mark O , the mid-point as one end of axis. Similarly, on the back face locate P and join OP by chain line to indicate the axis.
- Convert the hidden edge of the cylinder (half of the back side) to short dashes. Name the axis using capitals, and print the given dimensions as permitted for pictorial views (Refer to Fig. 16.10) to complete the drawing.

Example 16.4

Draw the isometric projection of a pentagonal prism of side of base 30 mm and height 60 mm, resting upon its base on HP and a rectangular face is parallel to VP.

Refer to Fig. 16.17.

- Draw the multiview projections of the pentagonal prism in the given position.
- Enclose the object into a rectangular box and name the corners.
- Construct the isometric scale and draw the isometric projection of the box using isometric lengths. Also add the symbol for isometric axes.
- Construct the isometric pentagon in the bottom

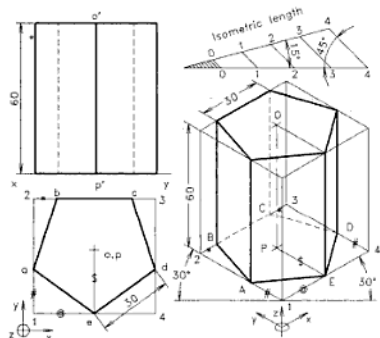


Fig. 16.17 Isometric projection (box method).

isometric plane, using the top view lengths converted into the isometric lengths.

- Draw the parallel isometric pentagon on the top isometric plane following the same procedure and join the corners by vertical lines. Also indicate the axis OP as done in Example 16.1.
- Convert the hidden edges to short dashes, finish the view and enter the dimensions as per BIS to complete the drawing.

Example 16.5

A hexagonal pyramid of height 50 mm and side 24 mm is resting on HP, keeping its axis vertical and one edge of the base parallel to VP. Draw isometric view of the solid.

Refer to Fig. 16.18.

- Draw the top and front views of the hexagonal pyramid in the given position and enclose the top view of in a rectangle.
- Construct the isometric view of the rectangle and draw the isometric base of the hexagonal pyramid in side. Also add the symbol for isometric axes.
- Locate the centre of the isometric hexagon and draw the vertical axis OP of given length. Then join the apex O to the six base corners to complete the pyramid.
- Finish the view using proper line types, indicate the hidden portion pyramid and dimension the figure as per BIS to complete the drawing.

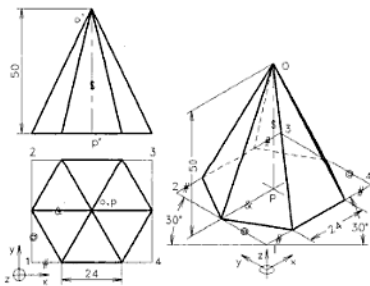


Fig. 16.18 Isometric view (coordinate method).

Example 16.6

A cone of height 50 mm and diameter 48 mm is resting on HP, keeping its axis vertical. Draw isometric view of the solid.

Refer to Fig. 16.19.

1. Draw the top and front views of the cone and enclose the top view in a square.
2. Construct the isometric view of the square and draw the isometric circle (ellipse) of the cone in side (Refer to Fig. 16.13). Also add the symbol for isometric axes.
3. Locate the centre of the isometric square and draw the vertical axis OP of given length. Then draw the outermost generators of cone by drawing tangents from O to the base (ellipse).
4. Finish the view using proper line types, indicate the hidden portion of the ellipse and dimension the figure as per BIS to complete the drawing.

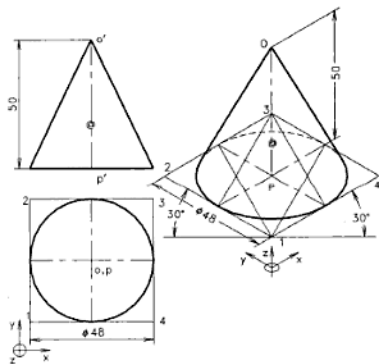


Fig. 16.19 Isometric view of solids (coordinate method).

Example 16.7

A frustum of a cone of base diameter 50 mm, top diameter 30 mm, and height 45 mm is resting upon its base on HP. Draw the isometric projection of the frustum.

Refer to Fig. 16.20.

1. Draw the top and front views of the frustum of cone and enclose the circles of top view inside squares.

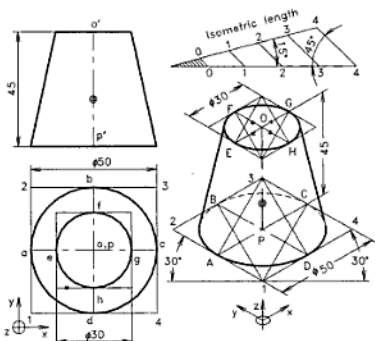


Fig. 16.20 Isometric projection of a frustum (coordinate method).

2. Construct the isometric scale and draw the isometric square 1234 for the base of the cone. Also add the symbol for isometric axes.
3. Construct the isometric circle (ellipse) ABCD inside the isometric square 1234 (refer to Fig. 16.13) and draw the vertical axis OP at the centre of it. Mark the height of the axis after converting it to the isometric length.
4. To construct the top isometric circle about the centre O, draw 30° inclined lines EG and FH through O and mark the length equal to the isometric radius of the top face of cone on them, i.e. $OE = OG = OF = OH$. Draw the isometric square 5678 (rhombus) parallel to these lines as given in figure and construct the isocircle (ellipse) EFGH inside.
5. Draw the outermost generators of the cone as tangents to the two ellipses. Note that all the true lengths should be multiplied by the isometric scale (82%) before marking, since an isometric projection is the requirement.
6. Finish the view using proper line types, indicate the hidden portion of the ellipse and dimension the figure as per BIS to complete the drawing.

Example 16.8

A square pyramid, edge of base 40 mm and axis 60 mm long, is lying on one of its triangular faces upon HP and its axis

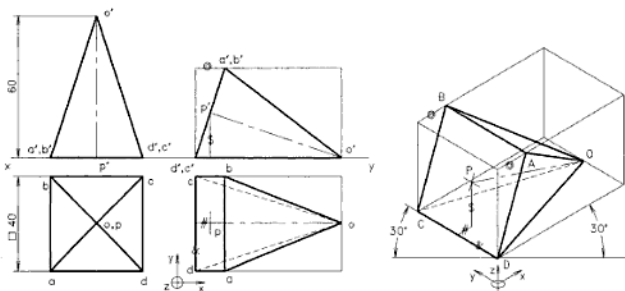


Fig. 16.21 Isometric view of a square pyramid (box-method).

parallel to VP. Draw the isometric view of the given pyramid showing the base.

Refer to Fig. 16.21.

1. Draw the front and top views of the pyramid in the given position and enclose it in a rectangular box.
2. Draw the isometric view of the box using the lengths measured along xy directions. Also add the symbol for isometric axes.
3. Mark the base points ABCD on the box corners and edges and locate the apex O at the middle of bottom edge of the back face. Join these points using thin lines to get the view.
4. To draw the inclined axis OP, locate the end P by marking the xyz distances obtained from the top and front views as shown in the figure. Join OP by chain line to indicate the axis. A short cross mark may be shown at P for which the lines are drawn parallel to the base edges.
5. Finish the view using proper line types to complete the drawing.

16.7 SECTIONED SOLIDS

Isometric view of a solid, sectioned by a cutting plane, can be drawn by box method or coordinate method. In box method, after drawing the isometric view of the box, the cutting plane is marked in it and the cut surface is drawn on that plane. But in coordinate method, the sectioned surface is obtained by marking the coordinates of the boundary points from an isometric plane as reference. Figure 16.22 shows isometric projection of a sectioned vertical pentagonal prism using coordinate method. Similarly, Fig. 16.23 shows isometric

view of a sectioned horizontal hexagonal prism. Generally coordinate method is found more easy to get the isometric projection or view of a sectioned solid.

Example 16.9

A pentagonal prism of side of base 30 mm and height 60 mm is resting on its base upon HP, keeping one base edge parallel and nearer to VP. The prism is cut by a section plane, 30° inclined to HP and passing through a point on the axis, 40 mm above the base. Draw isometric projection of the prism showing the sectioned surface.

Refer to Fig. 16.22.

1. Draw the top and front views of the sectioned prism in the given position and name the section as 12345. Enclose the plan in a rectangle.
2. Construct the isometric scale and draw the isometric projection of the rectangle to enclose the base of the prism. Also add the symbol for isometric axes.
3. Mark the base points ABCDE on the isometric rectangle and locate the axis end P at the middle of base by measuring the distances along xy directions and multiplying by 0.82. Join these points using thin lines to get the isometric pentagon.
4. To draw the sectioned prism, measure the distance $a'1'$, multiply by 0.82 and mark it from the corner A in the z direction to obtain the vertical edge A1. Similarly, mark the remaining four vertical edges and the axis as shown in figure. Join the corners to get the projection.
5. Finish the view using proper line types and hatch the cut surface to complete the drawing.

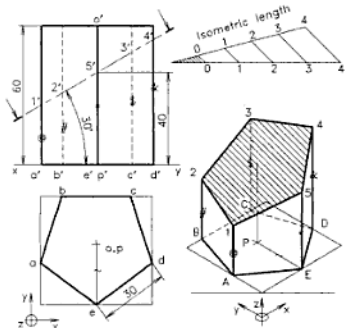


Fig. 16.22 Isometric projection (coordinate method).

Example 16.10

A hexagonal prism of side of base 20 mm and length 50 mm is resting on one of its rectangular faces upon HP, keeping the base parallel to VP. The prism is cut by a vertical section plane 30° inclined to VP and passing through the midpoint on the axis. Draw isometric view of the prism showing the sectioned surface.

Refer to Fig. 16.23.

1. Draw the front and top views of the sectioned prism in the given position and name the section as 123456. Enclose the elevation in a rectangle.

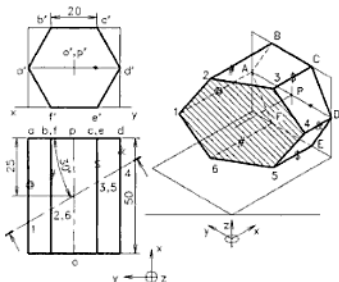


Fig. 16.23 Isometric view (coordinate method).

2. Draw the isometric view of the rectangle enclosing the plan and then construct the isometric view of the rectangle enclosing the elevation on it as shown in figure. Also add the symbol for isometric axes.
3. Mark the base points ABCDEF on the vertical isometric rectangle and locate the axis end P at the middle of base by measuring the distances along y and z directions. Join these points using thin lines to get the isometric hexagon.
4. To draw the sectioned prism, measure the distance a to 1, and mark it from the corner A in the x direction to obtain the horizontal edge A1. Similarly, mark the remaining five horizontal edges and the axis as shown in the figure. Join the corners to get the view.
5. Finish the view using proper line types and draw section lines on the cut surface to complete the drawing.

Example 16.11

A square pyramid of side of base 40 mm and height 60 mm is resting on its base upon HP, keeping the base edges equally inclined to VP. The pyramid is cut by a section plane, 30° inclined to HP and passing through the midpoint of the axis. Draw isometric view of pyramid showing the section.

Refer to Fig. 16.24.

1. Draw the top and front views of the sectioned pyramid in the given position and name the sectioned surface as 1234.

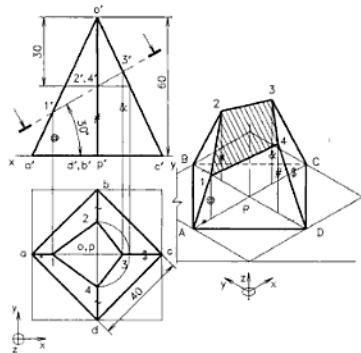


Fig. 16.24 Isometric view (coordinate method).

- Enclose the base $abcd$ in a square as shown in the figure. Draw the isometric view of this square and mark the base points $ABCD$ on it.
- To locate the corners 1, 2, 3, and 4 of the cut-surface, use the coordinate method. To obtain point 1, measure the coordinate values of the point in the x , y and z directions from the top and front views respectively and mark the same along the isometric axes. Similarly locate 2, 3 and 4, and join them by straight lines to get the cut-surface.
- Join the points 1234 to the respective base corners A, B, C and D to get the base portion of the pyramid. Locate the axis position P and draw the vertical axis through the point.
- Finish the view using proper line types, and hatch the cut surface to complete the drawing.

Example 16.12

A cylinder of diameter 48 mm base and 60 mm height, is resting upon its base on HP. A section plane of 45° inclination to HP bisects the axis of the cylinder. Draw the isometric view of the cylinder showing the sectioned surface.

Refer to Fig. 16.25.

- Draw the top and front views of the cylinder and mark the section plane at 45° . Enclose the base of the cylinder in a square.
- Construct the isometric view of the square and draw an isometric circle (ellipse) inside to represent the base of the cylinder.
- In the top view, draw diagonals to the square and mark the points 1, 2, 3, ..., 8 at the eight intersection

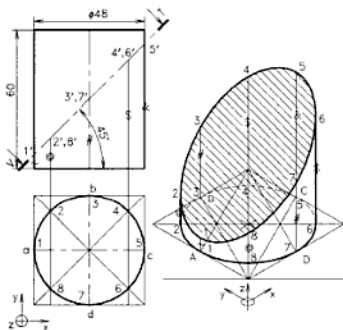


Fig. 16.25 Isometric view (coordinate method).

points of the diagonals on the circle. Mark these two diagonals in the isometric view of the square and locate the points 1, 2, 3, ..., 8 at the intersection of isocircle in the same order.

- To construct the isometric view of the sectioned cylinder, join the points 8 and 2 in the top view, and then produce to the front view to get point $2'8'$ on the section line. This vertical line through $2'8'$ is a generator of the cylinder in the front view. Similarly, mark generators through $3'7'$ and $4'6'$ on the section line. Points $1'$ and $5'$ represent the extreme points on the section line.
- Measure the heights (z values) of these generators from the base of cylinder to the points 1, 2, 3, ..., 8, in the front view and mark them in the same order on the isometric view as 1-1, 2-2, 3-3, ..., 8-8 in the z direction as shown in the figure. Join all the top points by a smooth curve to get the elliptical sectioned surface 1, 2, 3, ..., 8. Also draw vertical lines tangential to the two ellipses to represent the outermost generators of the cylinder.
- Finish the view using proper line types, and hatch the cut surface to complete the drawing.

16.8 COMBINATION OF SOLIDS

Drawing procedure for the isometric projection of a combination of two or more solids is similar to that of individual solids. The point to be specially considered is the relative position of them in the isometric view. Figure 16.26 shows the isometric view of a cone placed over a square slab.

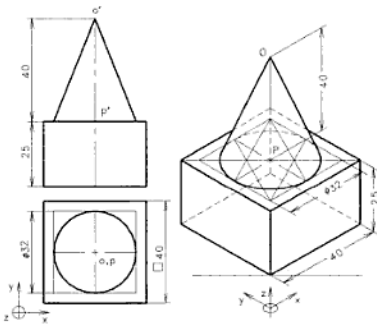


Fig. 16.26 Isometric view (combination of solids).

Here, the base of the cone is placed over the top face of the slab, so that the axes are coinciding.

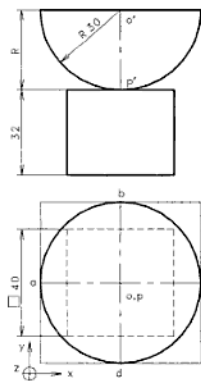
When a sphere is to be drawn in combination with the another solid, care should be taken to avoid mistakes in marking the centre of the sphere. The method of drawing the isometric projection and view of a hemisphere resting on a square slab, is given Fig. 16.27.

Example 16.13

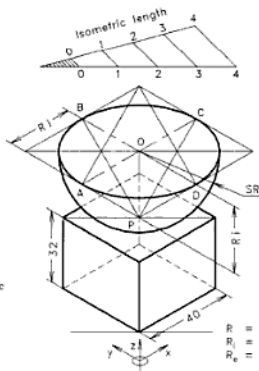
A cone of diameter 32 mm base and 40 mm height is surmounted over a square slab of 40 mm side and 25 mm thickness on HP so that one edge of the square is parallel to VP. Draw isometric view of the combination.

Refer to Fig. 16.26.

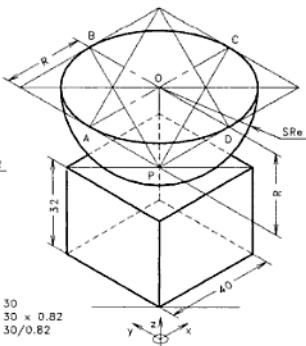
1. Draw the top and front views of the combination of solids keeping the cone centrally over the square slab. Enclose the base circle of the cone in a square.
2. Construct the isometric view of the square slab and mark point P at the middle of the diagonal of top face of the slab, as the axis end of cone. Draw an isometric square (rhombus) of 32 mm side keeping the centre at P. Construct isometric circle (ellipse) inside the rhombus to represent the base of the cone.



(a) Multiview projection



(b) Isometric projection



(c) Isometric view

Fig. 16.27 A hemisphere on a square slab.

3. Mark the axis height OP for the cone and complete the cone by drawing tangents to the ellipse.
4. Finish the view using proper line types, indicate the hidden portions and dimension the figure as per BIS to complete the drawing.

Isometric Projection of a Hemisphere Resting on a Slab

If a hemisphere with centre O and radius R is placed on a square slab as shown in Fig. 16.24(a), the line joining the centre of the sphere o' to the point of contact of the spherical surface with the top of slab p' will be vertical in the front view. If the sphere is tilted and brought to the isometric position, the vertical distance $o'p'$ will be inclined $35^\circ 16'$ and hence it will get foreshortened to 82% approximately as explained earlier. Thus, in the isometric projection of a hemisphere, the centre will be at a height of 82% of R from the point of contact. Here, OP is the isometric distance of $o'p'$ and is equal to $R_i = 82\% \text{ of } R$ [Fig. 16.27(b)]. However, the radius of the outer surface of the hemisphere is same as SR (because there is no difference in size if a sphere is tilted) and is equal to half the major axis of the ellipse representing the top circular plane.

Isometric View of a Hemisphere Resting on a Slab

In the preparation of isometric view of an object, the foreshortening of the isometric axes is ignored and the true length of the edges of the object are used directly for drawing. As a result, the isometric view obtained will be 22.5% larger in size. Similarly, if the isometric view of the hemisphere is drawn, the view obtained should be 22.5% larger than the given sphere itself [Fig. 16.27(c)]. So, the radius of the hemisphere in the isometric view should be enlarged to 22%, i.e. SR_e is equal to $1.225 \times R$, which is half the major axis of the ellipse representing the top circular plane. As the dimensions along the isometric axes are not foreshortened, the line joining the centre of the hemisphere to the point of contact of the spherical surface with the top of the slab $o'p'$ is also not foreshortened. Hence, $o'p' = OP = R$.

Hence, by the above explanations it is clear that:

1. For an isometric projection, all dimensions along xyz directions are foreshortened to 82% but the radius for the spherical surface should not be reduced.
2. For an isometric view, all dimensions along xyz directions are directly used but the radius for the spherical surface should be enlarged to 122.5%.

Example 16.14

A hemisphere of radius 30 mm is placed centrally on a square slab of side 40 mm and thickness 32 mm so that the flat circular surface is on the top. Draw the isometric projection and view of the solids in the given position.

Refer to Fig. 16.27.

1. Draw the multiview projection of the slab and hemisphere in the given position and enclose the circle in a square.
2. Construct the isometric scale and draw the isometric projection of the slab using isometric (82%) lengths.
3. Also draw the isometric view of the slab using the given dimensions directly.
4. To draw the isometric projection of the hemisphere, (b) locate the centre P of the top face of the isometric projection of the slab and draw a vertical line $OP = R$, i.e. the isometric length of $o'p'$ (30×0.82 mm). With centre O, draw the isometric square (rhombus) and construct the isocircle (ellipse) of radius R_e inside as shown in the figure. Construct a semicircle of radius equal to half the major axis of the ellipse (SR) at centre O to represent the spherical surface of the hemisphere. Note that the circle is not passing through the point P.

5. To draw the isometric view of the hemisphere, (c) locate the centre P of the top face of the isometric view of the slab and draw a vertical line $OP = R$, i.e. the length of $o'p'$ (30 mm). With centre O, draw the isometric square (rhombus) and construct the isocircle (ellipse) of radius R inside as shown in figure. Construct a semicircle of radius equal to half the major axis of the ellipse ($SR_e = 30/0.82$) at centre O to represent the spherical surface of the hemisphere. Note that here also the circle is not passing through the point P.
6. Finish the view and print the given dimensions to complete the drawing.

Example 16.15

A sphere of 18 mm radius is placed centrally over a hexagonal slab of side length 24 mm and thickness 25 mm. Draw isometric view of the combination.

Refer to Fig. 16.28.

1. Draw the multiview projection of the hexagonal slab and sphere in the given position. Enclose the hexagon in a square.
2. Construct the isometric view of the hexagonal slab using the given dimensions directly as shown.
3. To draw the isometric view of the sphere, locate the centre P of the top face of the isometric view of the slab (centre of line 1-4) and draw a vertical line $OP = R$, i.e. the length of $o'p'$ (18 mm). Construct a circle of radius $SR_e = 18/0.82$ mm at centre O to represent the surface of the sphere.
4. Finish the view and print the given dimensions to complete the drawing.

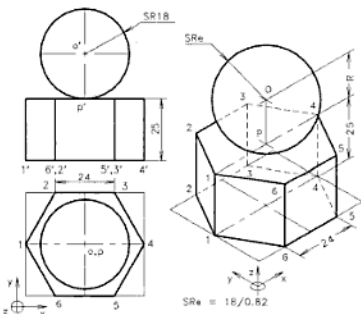


Fig. 16.28 Isometric view .

16.9 OBJECTS

Simple engineering objects can be very clearly described by isometric projection or view. Since the preparation of isometric view is simpler, generally that is preferred for engineering objects. The hidden lines are shown only if they are necessary. The following examples describe the procedure of drawing and they are self-explanatory. The construction lines are retained in order to understand the method of drawing. Since the problems are given as multiview projection, the student has to develop the capacity to read and visualize the shape of object, before starting the drawing. The location of the origin of the isometric axes has to be fixed in such a way that it brings out maximum information of the object.

Example 16.16

Draw the isometric view of the block shown in Fig. 16.29(a).

Refer to Fig. 16.29(b).

1. Locate the origin of the isometric axes on the left bottom corner of the top view and draw the isometric view of the box containing the block.
2. Mark the corners of the block relative to the edges of the box as shown in the figure.
3. Finish the view and print the given dimensions to complete the drawing.

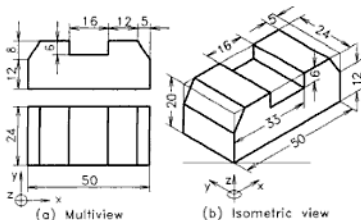


Fig. 16.29 A block.

Example 16.17

Draw the isometric view of the bracket shown in Fig. 16.30(a).

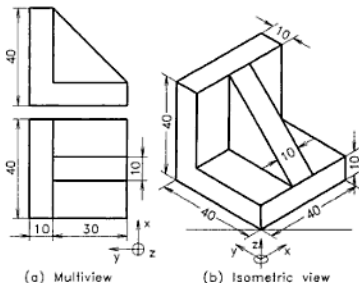


Fig. 16.30 A bracket.

Refer to Fig. 16.30(b).

1. Locate the origin of the isometric axes on the right bottom corner of the top view in order to get clear view of the web on the right isometric plane.
2. Draw isometric view of the bracket considering it as three slabs such as a square horizontal slab, a rectangular vertical slab and a vertical triangular slab joined together.
3. Finish the view and print the given dimensions to complete the drawing.

Example 16.18

Draw the isometric view of the machine part shown in Fig. 16.31(a).

Refer to Fig. 16.31(b).

Draw isometric view of the machine part considering it as three slabs such as a rectangular horizontal slab with a slot, a rectangular vertical slab and a semicircular vertical slab joined together.

Example 16.19

Draw the isometric view of the block shown in Fig. 16.32(a) and need not dimension.

Refer to Fig. 16.32(b).

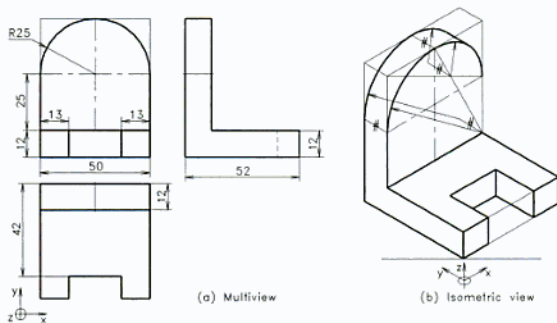


Fig. 16.31 A machine part.

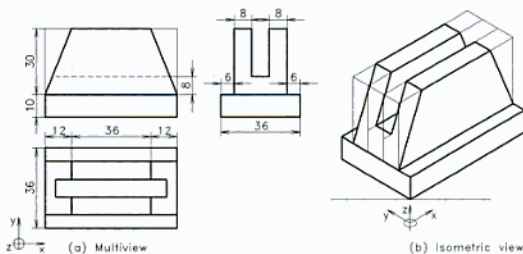


Fig. 16.32 A block.

Tools to solve isometric projection problems

1. The isometric axes in x and y directions are drawn at 30° inclination to horizontal and z axis in vertical direction.
2. In isometric projection, the dimensions measured from multiview along xyz directions should be foreshortened and marked in the corresponding isometric axes.
3. In isometric view, the dimensions measured from multiview along xyz directions are marked directly in the corresponding isometric axes.
4. The radius of arc representing a spherical surface should not be foreshortened in isometric projection, whereas in isometric view it has to be enlarged to 122.5%.

5. Box method or coordinate method can be used interchangeably in isometric drawings. For pyramids, cones and sectioned solids coordinate method is found more fast to reach solution whereas the box method is better for solids in inclined position or objects have intricate shapes.
6. A circle is seen as ellipse and it can be drawn by four centre method.
7. If the origin of axes is kept at the left bottom corner of the top view; the front, top and left side views are seen on right, top and left isometric planes respectively. Similarly, If the origin of axes is kept at the right bottom corner of the top view; the front, top and right side views are seen on left, top and right isometric planes respectively.

EXERCISES

SECTION A

(# Problems similar to workedout examples)

Solids

1. Draw the isometric projection of a rectangular prism of side of base $40\text{ mm} \times 24\text{ mm}$ and height 50 mm , resting upon its base on HP and the 40 mm long edges are parallel to VP. (#)
2. Draw isometric view of a pentagonal prism of 52 mm height and side 26 mm long, lying on HP with the axis perpendicular to VP. Select the origin of the isometric axes suitable to get the front view on the left isometric plane. (#)
3. Draw isometric view of a cylinder of 60 mm height and diameter 44 mm , lying on one of its generators on HP with the axis perpendicular to VP. Select the origin of the isometric axes suitable to get the front view on the right isometric plane. (#)
4. Draw the isometric projection of a hexagonal prism of side of base 26 mm and height 64 mm , resting upon its base on HP and a rectangular face is parallel to VP. (#)
5. A pentagonal pyramid of height 60 mm and side 28 mm is resting on HP, keeping its axis vertical and one edge of the base parallel to VP. Draw isometric view of the solid. (#)
6. A cone of height 70 mm and diameter 52 mm is resting on HP, keeping its axis vertical. Draw isometric view of the solid. (#)
7. A frustum of a cone of base diameter 56 mm , top diameter 32 mm , and height 52 mm is resting upon its base on HP. Draw the isometric projection of the frustum. (#)
8. A square pyramid, edge of base 52 mm and axis 64 mm long, is lying on one of its triangular faces upon HP and its axis parallel to VP. Draw the isometric view of the given pyramid showing the base. (#)
9. A pentagonal pyramid, edge of base 30 mm and axis 60 mm long, is lying on one of its triangular faces upon HP and its axis parallel to VP. Draw the isometric view of the given pyramid without showing the base surface.

Sectioned solids

10. A hexagonal prism of side of base 26 mm and height 64 mm is resting on its base upon HP, keeping one base edge parallel and nearer to VP. The prism is cut by a section plane, 30° inclined to HP and passing through a

point on the axis, 40 mm above the base. Draw isometric projection of the prism showing the sectioned surface. (#)

11. A pentagonal prism of side of base 24 mm and length 64 mm is resting on one of its rectangular faces upon HP, keeping the base parallel to VP. The prism is cut by a vertical section plane 30° inclined to VP and passing through the mid-point on the axis. Draw isometric view of the prism showing the sectioned surface. (#)
12. A hexagonal pyramid of side of base 26 mm and height 64 mm is resting on its base upon HP, keeping two base edges parallel to VP. The pyramid is cut by a section plane, 30° inclined to HP and passing through the midpoint of the axis. Draw isometric view of pyramid showing the section. (#)
13. A cylinder of diameter 56 mm base and 80 mm height is resting upon its base on HP. A section plane of 45° inclination to HP bisects the axis of the cylinder. Draw the isometric view of the cylinder showing the sectioned surface. (#)
14. A cylinder of diameter 50 mm base and 70 mm height is resting upon its base on HP. A section plane of 60° inclination to HP cuts the axis of the cylinder at a height of 55 mm from the base. Draw the isometric view of the cylinder showing the sectioned surface.

Combination of solids

15. A hexagonal pyramid of base edge 20 mm and height 50 mm is surmounted over a square slab of 50 mm side and 30 mm thickness on HP so that one side of the square and one base edge of the pyramid are parallel to VP. Draw isometric view of the combination. (#)
16. A hemisphere of radius 32 mm is placed centrally on a square slab of side 36 mm and thickness 30 mm so that the flat circular surface is on the top. Draw the isometric projection and view of the solids in the given position. (#)
17. A sphere of 20 mm radius is placed centrally over a pentagonal slab of side length 30 mm and thickness 36 mm . Draw isometric view of the combination. (#)

Objects

18. Draw isometric view of the block shown in Fig. 16.33. (#)
19. Orthographic views of a block are shown in Fig. 16.34. Draw the isometric view. (#)

47. Orthographic view of a V-block is shown in Fig. 16.45. Draw the isometric view.

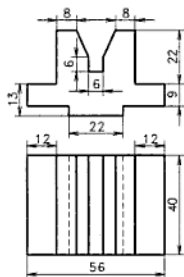


Fig. 16.45

48. A cast iron block is shown in front and side views. Draw its isometric view. Refer Fig. 16.46.

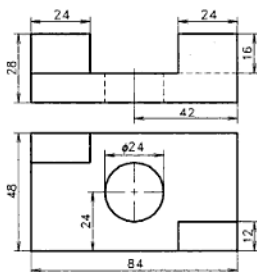


Fig. 16.46

49. A rod support is shown in Fig. 16.47. Draw its isometric view.

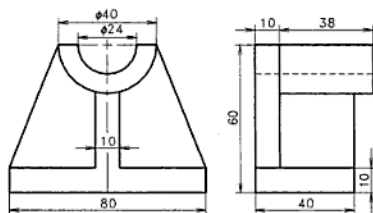


Fig. 16.47

50. Multiview projection of a crank is shown in Fig. 16.48. Draw the isometric view of the crank.

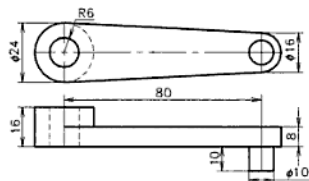


Fig. 16.48